

Decision Tree

Random Forest

Decision Tree Example

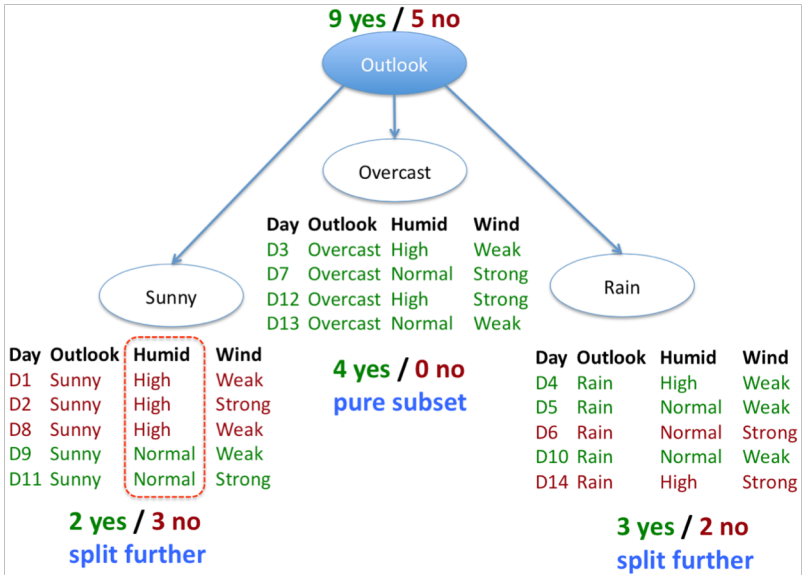
Training examples: 9 yes / 5 no

Day	Outlook	Humidity	Wind	Play
D1	Sunny	High	Weak	No
D2	Sunny	High	Strong	No
D3	Overcast	High	Weak	Yes
D4	Rain	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	Strong	No
D7	Overcast	Normal	Strong	Yes
D8	Sunny	High	Weak	No
D9	Sunny	Normal	Weak	Yes
D10	Rain	Normal	Weak	Yes
D11	Sunny	Normal	Strong	Yes
D12	Overcast	High	Strong	Yes
D13	Overcast	Normal	Weak	Yes
D14	Rain	High	Strong	No

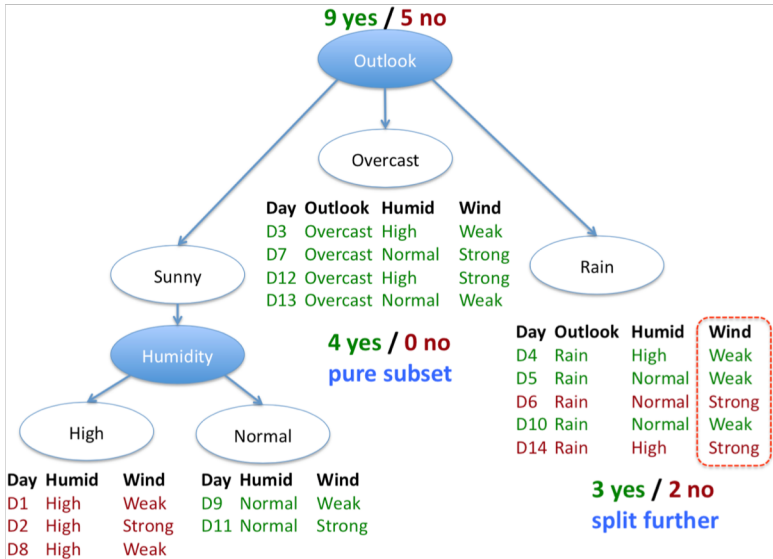
New data:

D15	Rain	High	Weak	?
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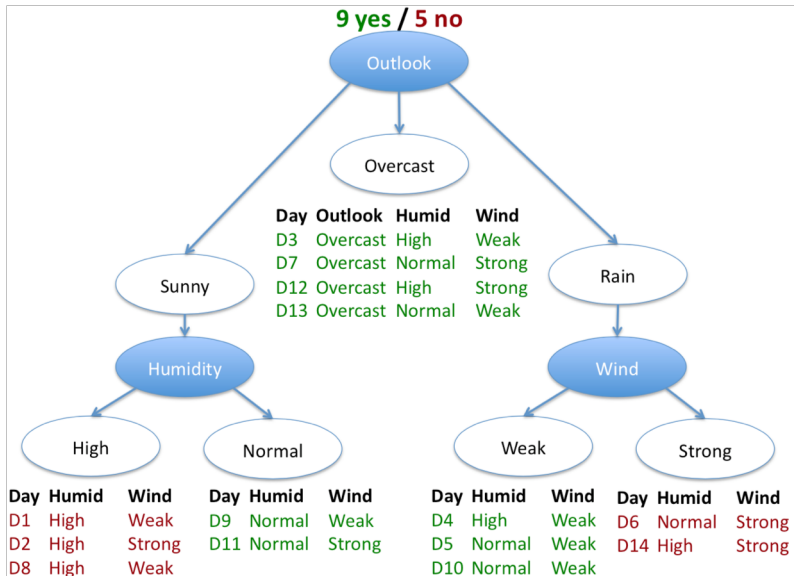
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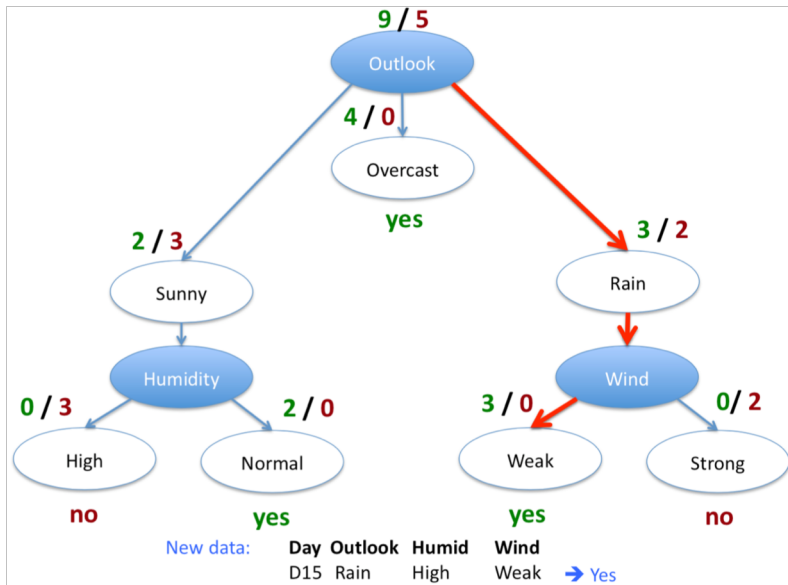
Decision Tree Example



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Decision Tree Example



ID3 Algorithm

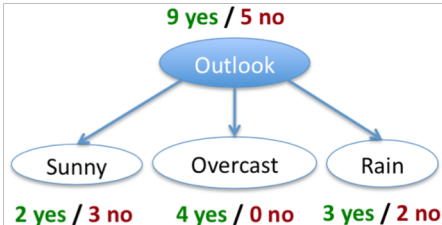
node = Root

examples = Training Set

Split (**node**, {**examples**}):

1. Find A, the **best attribute** for splitting the {**examples**}
2. Create decision nodes for attribute A (i.e., **child nodes** of **node**)
3. Split training {**examples**} to **child nodes**
4. If examples perfectly classified (subset is pure): STOP
else: iterate over new child nodes
 Split (**child_node**, {**subset of examples**})

Find The Best Attribute



- Want to measure “purity” of the split
 - more certain about Yes/No after the split
 - pure set (4 yes / 0 no) => completely certain (100%)
 - impure (3 yes / 3 no) => completely uncertain (50%)
 - can’t use $P(\text{“yes”} \mid \text{set})$:
 - must be symmetric: 4 yes / 0 no as pure as 0 yes / 4 no

Entropy

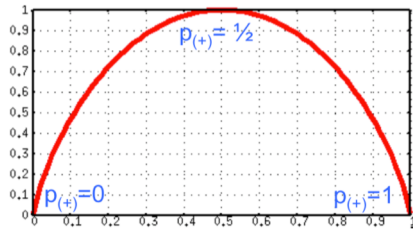
- Entropy: $H(S) = -p_{(+)} \log_2 p_{(+)} - p_{(-)} \log_2 p_{(-)}$ bits
 - S ... subset of training examples
 - $p_{(+)} / p_{(-)}$... % of positive / negative examples in S
- Interpretation: assume item X belongs to S
 - how many bits need to tell if X positive or negative

- impure (3 yes / 3 no):

$$H(S) = -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1 \text{ bits}$$

- pure set (4 yes / 0 no):

$$H(S) = -\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} = 0 \text{ bits}$$



Information Gain

- Want many items in pure sets
- Expected drop in entropy after split:

$$\text{Gain}(S, A) = H(S) - \sum_{V \in \text{Values}(A)} \frac{|S_V|}{|S|} H(S_V)$$

V ... possible values of A
 S ... set of examples $\{X\}$
 S_V ... subset where $X_A = V$

- Mutual Information
 - between attribute A and class labels of S

$$\begin{aligned}\text{Gain}(S, \text{Wind}) &= H(S) - \frac{8}{14} H(S_{\text{weak}}) - \frac{6}{14} H(S_{\text{strong}}) \\ &= 0.94 - \frac{8}{14} * 0.81 - \frac{6}{14} * 1.0 \\ &= 0.049\end{aligned}$$

